WASP-12b

R-0193

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-12_250114` · created 2026-07-16T21:12:51+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460689.6823 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.059 +/- 0.03
Transit depth (Rp/Rs)^2	0.34 +/- 0.35 [%]
Orbital Inclination (inc)	85.22 +/- 2.83
Airmass coefficient 1 (a1)	1.01 +/- 0.014
Airmass coefficient 2 (a2)	-0.011 +/- 0.028
Scatter in the residuals of the lightcurve fit is	1.39 %
Best Comparison Star	#2 - [334, 180]
Optimal Method	PSF photometry
Transit Duration (day)	0.0935 +/- 0.0029

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.059 ± 0.03 vs 0.1178 (deviation 1.76σ)
Duration fit vs published	0.0935 d vs 0.125 d (deviation 9.77σ)
Mid-time O–C	-3.7 min
Depth SNR	1.8
Residual scatter	1.39 %

Light curve

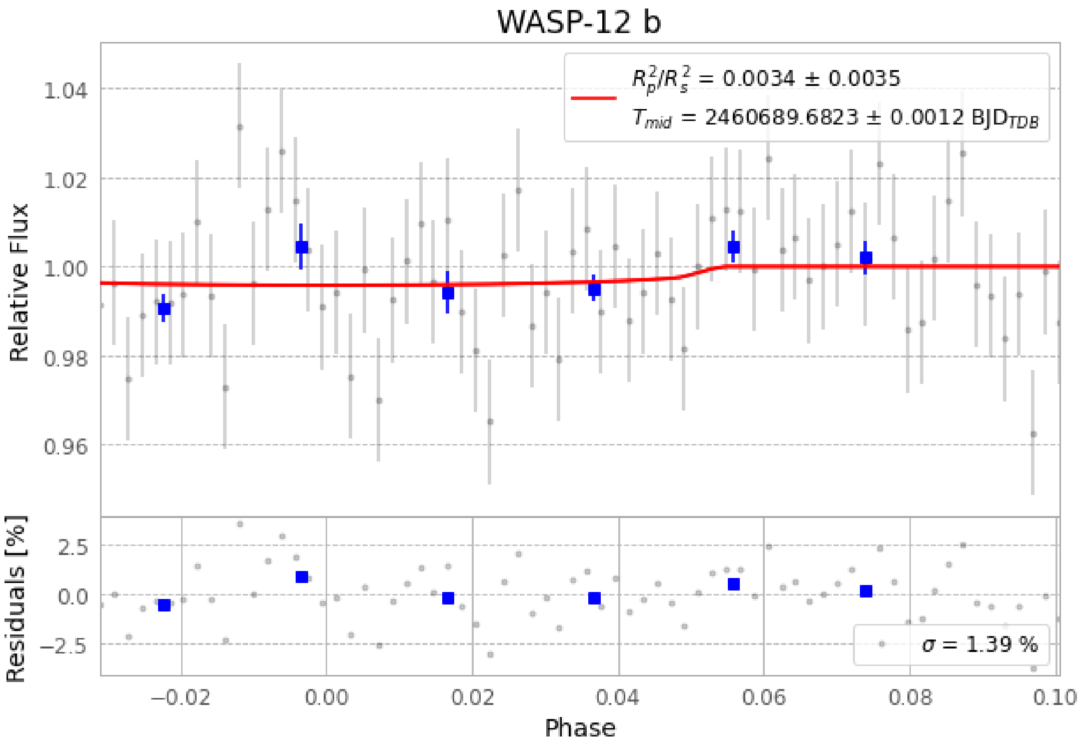

Figure 1 — FinalLightCurve_WASP-12 b_2025-01-13.png (SHA-256 19b1bf1b0660c0b3…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [446, 197], 2 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 ec38c434cc597a6e…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 378b18d`

Data provenance

71 FITS frames (47 MB), `WASP-12250114032115.FITS` → `WASP-12250114065115.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-250114.log (401a709e6b48…), FinalLightCurve_WASP-12 b_2025-01-13.png (19b1bf1b0660…), FinalLightCurve_WASP-12 b_2025-01-13.csv (512d317e6c6e…)

Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)